Information-Induced Conservatism Frontier of Distributed Stability Certificates

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Abstract

Conservatism in distributed stability certification is usually treated as a limitation of the certificates themselves. Therefore, to reduce the conservatism, researcher seeks richer Lyapunov functions, tighter relaxations, or more powerful optimization methods. But this view misses a more fundamental limitation: a distributed certificate only sees the information exposed by its reporting interface. Once that information is compressed, physically different systems may become indistinguishable. If a stable system and a less stable or unstable system produce the same report, no sound method can separate them using that report. The resulting conservatism is caused by missing information, not by the insufficient analysis methods. This paper asks where this fundamental information limit lies, and how much additional information is needed to move it.

We model the available information as an information structure that partitions admissible systems into information fibers. Within a fiber, every certificate is given identical information and therefore cannot distinguish between the different physical systems it contains. Accordingly, the strongest sound certificate for a fiber is limited by the smallest stability margin across all systems in that fiber.This gives a method-independent benchmark. It also separates total conservatism into two parts: irreducible information-induced conservatism and additional method-induced conservatism. The framework then characterizes information order, task sufficiency, and the stability value of additional information. Three network–node studies expose the underlying mechanism. Network-spectrum compression loses the alignment between network modes and heterogeneous nodes. Scalar or eigenvalue-only passivity reports lose local directional alignment. Independent local certificates lose cross-node relations. These examples show that the critical missing information is often relational, not dimensional.

We then move from diagnosis to information design. An information–conservatism frontier quantifies the smallest irreducible conservatism achievable under a prescribed budget. Its inverse problem gives the minimum information required for a desired conservatism. Through a task-relevant witness-pair characterization, we turn it into a weighted covering problem. For matrix passivity reports, we further derive an exact directional-information frontier. A local sensitivity law gives a computable approximation and identifies which unresolved quantities are most valuable to reveal. The resulting framework tells whether conservatism should be reduced by improving the certificate or by enriching the information.

1 Introduction

Distributed and compositional stability certificate is a recurring challenge in large-scale interconnected systems, where assembling and repeatedly analyzing a complete closed-loop model can be computationally expensive or impractical. Rather than reconstructing the entire model, distributed approaches seek local conditions whose satisfaction guarantees stability. Power-system stability studies provide prominent examples, including small-phase/passivity-based [1–3], small-gain-based [4], gain–phase mixed [5], and integral-quadratic-constraint-based methods [6]. The broader architectural idea is not specific to power systems: the certificate operates on selected component and interconnection descriptions rather than on the complete physical system. This compression makes scalable certificate possible, but it can also make

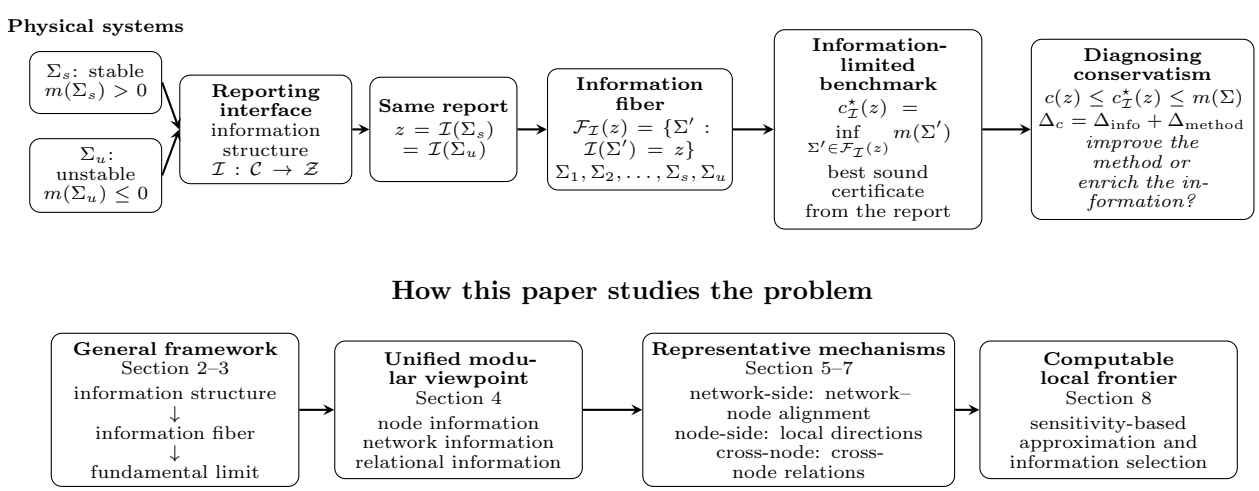


Figure 1: Overview of the information-limited view of distributed stability certification and the organization of this paper. A reporting interface induces an information structure under which physically different systems may produce the same report and therefore belong to the same information fiber. The best sound certificate attainable from that report is the worst-case stability margin over the fiber, providing the information-limited benchmark $c^\star_{\mathcal{I}}$. Comparison with a particular certificate separates total conservatism into information-induced and method-induced components and indicates whether further improvement should target the certificate method or the reporting interface. The lower panel summarizes the progression from the general framework to the modular interpretation, representative mechanisms of information-induced conservatism, and the computable local information frontier.

the resulting certificates conservative. *Two fundamentally different sources of conservatism must therefore be distinguished: the analysis method may fail to exploit the information it receives, or the available information itself may be insufficient for the stability question being asked.*

Much of the recent progress has focused on making the resulting certificates less conservative. Passivity restrictions have been relaxed through lifting the scalar passivity indices to matrix-valued descriptions to retain directional and inter-channel information [7,8]. Multiplier selection for broad classes of decentralized frequency-domain certificates has been formulated as a convex semidefinite program, enabling systematic searches for the least-conservative certificate within a prescribed family [9]. These developments address an important methodological question: *how effectively does a certificate use the information supplied to it?* They do not, however, answer the logically prior question: *how much can any sound certificate accomplish with that information in the first place?* In the extreme case, if a stable and an unstable system are indistinguishable under the available information, no improvement of the certificate algorithm alone can certify the stable one.

This distinction is operational rather than merely conceptual. When a distributed certificate is conservative, one may invest in a richer Lyapunov construction, a larger multiplier family, a tighter relaxation, or a more expensive optimization. Such effort is justified only if the existing information still permits a better sound certificate. If the best certificate compatible with the current reporting interface is already restrictive, algorithmic refinement cannot recover the missing margin. Instead, the reporting interface itself must be enriched. The central purpose of this paper is therefore to provide a benchmark that tells these two situations apart.

A simple example makes the issue concrete. Suppose a distributed certificate knows the individual node dynamics with the network Laplacian spectrum. Imagine a star-shaped interconnection: one hub is connected to all the other nodes. Suppose all subsystems have the same dynamics except one that with a stronger dynamic gain. Compare two cases: this high-gain subsystem is at the hub in the first

case and at a leaf in the second. Both networks are the same star and therefore have exactly the same network spectrum. But the high-gain subsystem interacts with the rest of the system very differently at the hub than at a leaf. For a suitable coupling strength and a class of node dynamics, one placement can remain asymptotically stable while the other fails the Hurwitz stability test. A certificate that knows the individual node models but receives only the network eigenvalues sees the same report in both cases, so it cannot soundly distinguish the stable system from its unstable counterpart. The missing information is not another eigenvalue; it is how the node dynamics are placed relative to the network's critical directions. This example illustrates a possible information insufficiency and will be further clarified in Section 5.

This distinction turns conservatism into an information-sufficiency problem. Blackwell's comparison of experiments provides a classical way of ordering information structures according to whether one can be obtained from another by garbling [10]. Alternatively, the data-informativity framework asks whether available data are sufficient for a specified analysis or control task over all systems consistent with those data [11, 12]. In distributed stability analysis, however, the relevant object is not necessarily a data set, but the information structure induced by the reporting interface itself. A network spectrum, a scalar or matrix-valued component index, or a collection of independent local descriptions may identify physically different interconnected systems as identical from the certificate's viewpoint. *The fundamental benchmark is therefore not another sufficient stability condition, but the best stability statement that can be made by any sound method restricted to the same information.* To the best of our knowledge, this information-limited optimum and the resulting separation between information-induced and method-induced conservatism have not been characterized for distributed stability certificates in the literature.

A related algorithm-independent viewpoint appears in information-based complexity, where the radius of information quantifies the irreducible worst-case error associated with restricted information structures [13]. The present setting differs in two aspects in our problem: soundness imposes a one-sided lower-certificate requirement, and the admissible information structures are dictated by modular network–node reporting interfaces rather than by arbitrary measurements. This motivates optimizing not over unrestricted structures, but over physically meaningful reporting-interface settings.

Throughout the paper, "information" is used in this task-specific distinguishability sense, rather than in the Shannon sense. A higher-dimensional report can have little or even zero stability value if it does not separate the systems that determine the worst case. Mathematically, evaluating the best certificate over an information fiber resembles a robust worst-case calculation, but the object of study is different: robust analysis typically starts from a prescribed uncertainty set, whereas here the information fiber is induced by the certificate information structure itself. The question is therefore not only how to analyze a fixed information fiber, but how the information structure refines that fiber and moves the fundamental certification limit.

This paper develops an information-based framework for answering this question. The framework is stated for general classes of interconnected systems and task-dependent stability margins; power-system certificate interfaces serve as motivating examples rather than a restriction on its scope. Its contributions follow a benchmark–diagnosis–design logic.

1. **Benchmark: the best certificate permitted by fixed information.** We define the point-wise largest sound certificate as the worst-case margin over all systems compatible with the reported information, providing a method-independent limit before a particular Lyapunov function, multiplier, or numerical algorithm is chosen.

2. **Diagnosis: separating information- and method-induced conservatism.** The benchmark decomposes total conservatism into information-induced conservatism and method-induced conservatism. Information sufficiency is characterized on the induced information fibers, with stable/unstable pairs in the same fiber serving as constructive impossibility witnesses.

3. **Design: identifying stability-relevant information.** We order information structures and define a budgeted information–conservatism frontier over admissible refinements. Network-, node-,

and cross-node examples expose the lost relations, while witness pairs, an exact passivity-direction frontier, and local sensitivity provide mechanisms for selecting useful additional information.

Together, these results turn conservatism analysis into a decision problem: *whether the next improvement should target the certificate method or the information structure.* Fig. 1 summarizes the central question and the organization of the paper.

The remainder of the paper is organized as follows. Section 2 introduces information structures and the optimal certificate under fixed information. Section 3 develops the corresponding equivalence-class, information-sufficiency, and fundamental-limit results. It further defines the budgeted information–conservatism frontier over admissible reporting-interface refinements and gives a witness-pair characterization of minimum sufficient information.Section 4 explains the network–node decomposition and unifies the sources of information-induced conservatism. Sections 5–7 then study network-side, node-side, and cross-node information-induced conservatism, respectively. Section 8 develops a computable local approximation of the information frontier from stability sensitivity, including a budgeted first-order information-selection law, and Section 9 concludes the paper.

Notation: The real and complex number sets are denoted by $\mathbb{R}$ and $\mathbb{C}$. For a matrix M, M^H denotes its Hermitian transpose, $\lambda(M)$ its eigenvalue, $\mathrm{spec}(M)$ its spectrum, and $\alpha(M) = \max \Re(\mathrm{spec}(M))$ the largest real part of its eigenvalues. The set of n-dimensional symmetric matrices is represented by $\mathbb{S}^n$.

2 Information Structure and Value for Stability Certificate

This section builds the concept framework required by the certificate question. We first specify the *ground-truth task margin* m, and then specify *what the distributed certificate can observe* through the information structure $\mathcal{I}$. A concrete method is represented by a certificate map c acting only on that report. Soundness compares c with the ground truth and leads to the information-limited optimum $c^\star_{\mathcal{I}}$, which is the best certificate attainable without changing the information structure. The gap between m and $c^\star_{\mathcal{I}}$ is then the irreducible information-induced conservatism Δ_{info}. Finally, ordering and refinement of information structures quantify how this limit changes when additional information is supplied. This sequence is the organizing line used throughout the remainder of the paper: *task → information → sound certificate → information-limited optimum → conservatism*

2.1 Stability Evaluation Task and Information Structure

Consider a class of admissible interconnected systems $\Sigma \in \mathcal{C}$, where Σ contains the complete information about the node dynamics, network topology, parameters, and operating point relevant to the stability task.

Definition 1 (Ground-Truth Stability Margin)**.** *A specified stability-evaluation task is represented by a real-valued map*

$$\boxed{m : \mathcal{C} \to \mathbb{R}, \qquad \Sigma \mapsto m(\Sigma)}, \tag{1}$$

where $m(\Sigma) > 0$ if and only if Σ is stable. The quantity $m(\Sigma)$ is called the ground-truth stability margin *of the complete system Σ.*

The particular choice of m is task-dependent. For small-signal stability, a natural choice is the negative closed-loop spectral abscissa. Other margins, such as the distance to a prescribed $\mathcal{D}$-stability boundary or an admissible coupling margin, can be used without changing the framework below.

Definition 2 (Information Structure and Information Fiber)**.** *A distributed or compositional stability analysis is assumed to observe the complete system only through an information structure*

$$\boxed{\mathcal{I} : \mathcal{C} \to \mathcal{Z}, \qquad z = \mathcal{I}(\Sigma)}, \tag{2}$$

where $\mathcal{I}$ is called the information structure. *For a reported observation $z \in \mathcal{Z}$, define the corresponding* information fiber *as*

$$\mathcal{F}_{\mathcal{I}}(z) := \left\{ \Sigma' \in \mathcal{C} : \mathcal{I}(\Sigma') = z \right\}. \tag{3}$$

When the report is generated by a particular system Σ, we use the shorthand

$$\mathcal{F}_{\mathcal{I}}(\Sigma) := \mathcal{F}_{\mathcal{I}}(\mathcal{I}(\Sigma)). \tag{4}$$

The information structure $\mathcal{I}$ may collect local node reports, network or operating-state reports, selected model coefficients, or any other quantities available to the certificate. All systems in $\mathcal{F}_{\mathcal{I}}(z)$ are indistinguishable to any method whose input is restricted to $\mathcal{I}$. Thus, $\mathcal{I}$ partitions the physical system class into information fibers, and stability certification is fundamentally a task over these fibers rather than over individual models. This view is closely related to data informativity [11]. The induced equivalence-class structure and its implications for fundamental conservatism are developed formally in Section 3.

Example 1 (Passivity-Based Stability Evaluation)**.** *Passivity provides a concrete example for the proposed methodology, which will be reused for illustration throughout the paper. In a standard passivity-based distributed assessment, each node is represented by a local passivity index, while the network is represented by a complementary description. A compositional interconnection condition then combines these local reports to certify the closed-loop system without requiring the certificate to reveal detailed node models. Let the matrix passivity index [8] of the i-th node be $\Pi_i \in \mathbb{S}^2$ and the network index be $\Pi_g \in \mathbb{S}^{2n}$. Define $\Pi = \mathrm{blkdiag}(\Pi_1, \ldots, \Pi_n)$; a typical condition then takes the form $\Pi - \Pi_g \succ 0$. Hence the passivity method operates on compressed information Π rather than on the complete system Σ.*

A legacy scalar reporting interface further compresses each matrix report to $\pi_i = \lambda_{\min}(\Pi_i)$ [14]. The scalar report merges all nodes sharing the same weakest-direction index. Matrix-valued passivity retains precisely part of the information discarded by this scalarization. Denote the scalar and matrix passivity information structures by $\mathcal{I}^{\Pi}_{\mathrm{sc}}$ and $\mathcal{I}^{\Pi}_{\mathrm{mat}}$, respectively.

2.2 Stability Certificates, Soundness, and Optimality

Definition 3 (Stability-Margin Certificate)**.** *For a fixed information structure $\mathcal{I}$, a particular stability-analysis method using only the reported information $z = \mathcal{I}(\Sigma)$ is represented by a map*

$$\boxed{c : \mathcal{Z} \to \mathbb{R}, \qquad z \mapsto c(z)}, \tag{5}$$

where $c(z)$ is the stability margin certified by that method, expressed in the same margin convention as m. A positive value $c(z) > 0$ certifies stability, whereas $c(z) \leq 0$ means only that the method cannot certify a positive margin from the available information.

Example 2. *The certificate c may be obtained from a spectral criterion, a passivity criterion, or any other stability-analysis method, provided that its output is expressed in the same stability-margin convention as m.*

The three basic objects introduced above have distinct roles: m evaluates the true margin from the complete system, $\mathcal{I}$ specifies what information is available to a distributed certificate, and c maps that available information to a certified margin. Equivalently, the complete and distributed evaluation paths are

$$\Sigma \xrightarrow{m} m(\Sigma), \qquad \Sigma \xrightarrow{\mathcal{I}} z \xrightarrow{c} c(z). \tag{6}$$

Definition 4 (Sound Stability-Margin Certificate)**.** *A certificate map $c : \mathcal{Z} \to \mathbb{R}$ is* sound *if*

$$c(\mathcal{I}(\Sigma)) \leq m(\Sigma), \qquad \forall \Sigma \in \mathcal{C}. \tag{7}$$

A sound certificate may underestimate the true margin, but it cannot overestimate it.

Theorem 1 (Optimal Certificate under Fixed Information Structure)**.** *For a fixed information structure $\mathcal{I}$, the point-wise largest sound certificate is*

$$\boxed{c^\star_{\mathcal{I}}(z) = \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma)}. \tag{8}$$

Proof. Let c be any sound certificate. For every $\Sigma \in \mathcal{F}_{\mathcal{I}}(z)$, (7) gives $c(z) \le m(\Sigma)$, and hence

$$c(z) \le \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma). \tag{9}$$

Conversely, the right-hand side of (8) does not exceed $m(\Sigma)$ for any system in the same information fiber and is therefore sound. Thus it is the point-wise optimum among all sound certificates. □

When evaluated at a particular system, we use the shorthand $c^\star_{\mathcal{I}}(\Sigma) := c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma))$. Theorem 1 immediately gives, for every sound certificate c,

$$c(\mathcal{I}(\Sigma)) \le c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) \le m(\Sigma). \tag{10}$$

Equation (10) is the central hierarchy of the framework: m is the oracle margin, $c^\star_{\mathcal{I}}$ is the best margin permitted by the available information, and c is the margin delivered by a particular sound method.

The significance of Theorem 1 is independent of the chosen analysis technique, such as a particular Lyapunov function, passivity condition, LMI, or numerical algorithm. *Even an arbitrarily powerful certificate cannot exceed $c^\star_{\mathcal{I}}$ without using information that further refines the current information fiber.*

For a particular sound method c, its point-wise conservatism is

$$\Delta_c(\Sigma; \mathcal{I}) := m(\Sigma) - c(\mathcal{I}(\Sigma)) \ge 0. \tag{11}$$

Definition 5 (Information-Induced Conservatism)**.** *For the true system Σ, the irreducible conservatism caused solely by the information structure $\mathcal{I}$ is*

$$\boxed{\Delta_{\text{info}}(\Sigma; \mathcal{I}) := m(\Sigma) - c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma))}. \tag{12}$$

For a particular sound method c, define its *method-induced conservatism* as

$$\Delta_{\text{method}}(\Sigma; \mathcal{I}) := c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) - c(\mathcal{I}(\Sigma)) \ge 0. \tag{13}$$

Combining (11)–(13) gives

$$\boxed{\Delta_c(\Sigma; \mathcal{I}) = \underbrace{m(\Sigma) - c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma))}_{\Delta_{\text{info}}(\Sigma;\mathcal{I})} + \underbrace{c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) - c(\mathcal{I}(\Sigma))}_{\Delta_{\text{method}}(\Sigma;\mathcal{I})}} \tag{14}$$

where both terms on the right-hand side are nonnegative. *Thus Δ_{info} is the information-induced conservatism that no sound method can remove without additional information, whereas Δ_{method} is the method-induced conservatism measuring how far a particular method is from the information-limited optimum.*

The hierarchy can be read as the short chain

$$\Sigma \xrightarrow{\mathcal{I}} z \xrightarrow[\text{best possible}]{c^\star_{\mathcal{I}}} c^\star_{\mathcal{I}}(z) \le m(\Sigma), \qquad c(z) \le c^\star_{\mathcal{I}}(z),$$

which separates two questions that are otherwise easy to conflate: whether the *information* is sufficient enough, and whether the *chosen method* uses that information well. The upper panel of Fig. 1 visualizes this hierarchy together with its decisive distinguishing failure: two systems on different sides of the stability boundary that collapse to the same report.

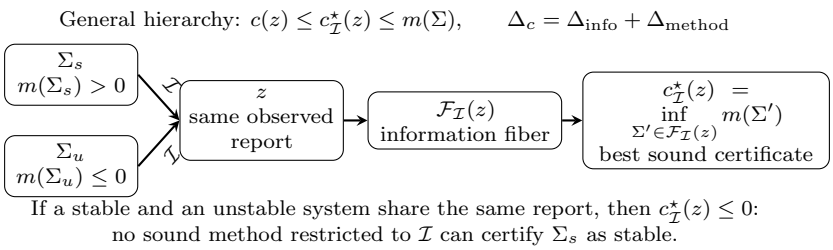


Figure 2: The information-limited benchmark and its basic failure witness. The information structure places all systems in $\mathcal{F}_{\mathcal{I}}(z)$ in the same information fiber, so the best sound certificate is determined by the worst margin in that fiber. If the fiber crosses the stability boundary, no refinement of the certificate algorithm alone can certify its stable member; the information structure must be refined. The pairwise statement is formalized in Corollary 1.

2.3 Information Order and Stability Value

Definition 6 (Deterministic Informativeness Order)**.** *If there exists a deterministic map f such that*

$$\mathcal{I}_1 = f \circ \mathcal{I}_2, \tag{15}$$

we write $\mathcal{I}_2 \succeq \mathcal{I}_1$ and say that $\mathcal{I}_2$ is at least as informative as $\mathcal{I}_1$ for distributed stability evaluations.

This is the deterministic counterpart of the Blackwell garbling order [10]. Strictly, $\succeq$ is a preorder on information structures. After quotienting out information structures that can be deterministically recovered from each other, it becomes a partial order on the resulting information structures. Moreover, if $\mathcal{I}_2 \succeq \mathcal{I}_1$, every $\mathcal{I}_2$-fiber is contained in the corresponding $\mathcal{I}_1$-fiber. Therefore, for any $\Sigma \in \mathcal{C}$,

$$c^\star_{\mathcal{I}_2}(\mathcal{I}_2(\Sigma)) \ge c^\star_{\mathcal{I}_1}(\mathcal{I}_1(\Sigma)),\ \Delta_{\text{info}}(\Sigma; \mathcal{I}_2) \le \Delta_{\text{info}}(\Sigma; \mathcal{I}_1). \tag{16}$$

Example 3 (Scalar-to-Matrix Passivity Refinement)**.** *Returning to Example 1, the scalar report is a deterministic function of the matrix report through the minimum-eigenvalue map. Therefore*

$$\mathcal{I}^{\Pi}_{\text{mat}} \succeq \mathcal{I}^{\Pi}_{\text{sc}}, \qquad \mathcal{F}_{\mathcal{I}^{\Pi}_{\text{mat}}}(\Sigma) \subseteq \mathcal{F}_{\mathcal{I}^{\Pi}_{\text{sc}}}(\Sigma). \tag{17}$$

For small-signal stability, choose the task margin $m(\Sigma) = m_\alpha(\Sigma) := -\alpha(\Sigma)$, where $\alpha(\Sigma)$ is the closed-loop spectral abscissa. Then

$$c^\star_{\mathcal{I}^{\Pi}_{\text{mat}}}(\mathcal{I}^{\Pi}_{\text{mat}}(\Sigma)) \ge c^\star_{\mathcal{I}^{\Pi}_{\text{sc}}}(\mathcal{I}^{\Pi}_{\text{sc}}(\Sigma)). \tag{18}$$

Thus, a reduction in conservatism from scalar to matrix passivity information can be interpreted without first selecting a particular matrix LMI: the matrix report removes some adverse directional systems that remain compatible with the scalar report. Whether a specific certificate c fully exploits this additional information is a separate question; Section 6 quantifies this information-induced conservatism explicitly.

If a refinement augments $\mathcal{I}$ by additional information $\mathcal{J}$, then $c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) \ge c^\star_{\mathcal{I}}(\Sigma)$.

Definition 7 (Stability Value of Information)**.** *For additional information $\mathcal{J}$, define its local value relative to $\mathcal{I}$ at Σ as*

$$\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) := c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) - c^\star_{\mathcal{I}}(\Sigma). \tag{19}$$

Proposition 1. *For additional information $\mathcal{J}$, the local value relative to $\mathcal{I}$ at Σ satisfies*

$$\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) = -(\Delta_{\text{info}}(\Sigma; \{\mathcal{I}, \mathcal{J}\}) - (\Delta_{\text{info}}(\Sigma; \mathcal{I})) \tag{20}$$

Proof. It is straightforward to substitute (19) into (12). □

The definition measures information in the same unit as the stability task. A high-dimensional report can have zero value if it removes no system that determines the fiber worst case, while one directional quantity can have substantial value if it separates the critical stable and unstable systems. *In other words, the amount of reported information need not indicate its value for a specific stability-evaluation task.*

3 Conservatism Limits from Perspective of Information Equivalence Class

3.1 Information-Induced Equivalence Classes

The information structure in Definition 2 does more than reduce the dimension of the available description: it partitions the physical system class according to what the certificate can and cannot distinguish. In other words, it leads to the informational indistinguishable system class.

Proposition 2 (Information-Induced Equivalence Relation and Quotient Space)**.** *For a fixed information structure $\mathcal{I} : \mathcal{C} \to \mathcal{Z}$, define the relation $\sim_{\mathcal{I}}$ on $\mathcal{C}$ as*

$$\Sigma_1 \sim_{\mathcal{I}} \Sigma_2 \quad \Longleftrightarrow \quad \mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_2). \tag{21}$$

Then $\sim_{\mathcal{I}}$ is an equivalence relation, and the corresponding quotient space

$$\mathcal{C}/\!\sim_{\mathcal{I}} = \{[\Sigma]_{\mathcal{I}} : \Sigma \in \mathcal{C}\} \tag{22}$$

is the space of information-induced indistinguishable system classes. Each equivalence class satisfies $[\Sigma]_{\mathcal{I}} = \mathcal{F}_{\mathcal{I}}(\Sigma)$.

Proof. Reflexivity follows from $\mathcal{I}(\Sigma) = \mathcal{I}(\Sigma)$. Symmetry follows because $\mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_2)$ implies $\mathcal{I}(\Sigma_2) = \mathcal{I}(\Sigma_1)$. Transitivity follows because $\mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_2)$ and $\mathcal{I}(\Sigma_2) = \mathcal{I}(\Sigma_3)$ imply $\mathcal{I}(\Sigma_1) = \mathcal{I}(\Sigma_3)$. Hence $\sim_{\mathcal{I}}$ is an equivalence relation. By definition, $[\Sigma]_{\mathcal{I}}$ contains exactly the systems with the same information representation as Σ, which is $\mathcal{F}_{\mathcal{I}}(\Sigma)$. □

Consequently, a method restricted to $\mathcal{I}$ operates on $\mathcal{C}/\!\sim_{\mathcal{I}}$, rather than on an individual physical realization. The stability task is therefore preserved exactly only when the information partition is sufficiently fine for that task.

3.2 Task Sufficiency and Within-Class Conservatism

By Theorem 1, the point-wise information-induced conservatism can be written directly on the equivalence class as

$$\boxed{\Delta_{\text{info}}(\Sigma; \mathcal{I}) = m(\Sigma) - \inf_{\Sigma' \in \mathcal{F}_{\mathcal{I}}(\Sigma)} m(\Sigma')}. \tag{23}$$

Thus, once the information structure is fixed, the irreducible conservatism is completely determined by the variation of the stability task within the corresponding equivalence class.

Theorem 2 (Equivalence-Class Width and Information-Induced Conservatism)**.** *For any reported value $z \in \mathcal{I}(\mathcal{C})$,*

$$\sup_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} \Delta_{\text{info}}(\Sigma; \mathcal{I}) = \sup_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma) - \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma). \tag{24}$$

Consequently, the worst-case information-induced conservatism of the information structure is

$$\overline{\Delta}_{\text{info}}(\mathcal{I}) := \sup_{\Sigma \in \mathcal{C}} \Delta_{\text{info}}(\Sigma; \mathcal{I}) = \sup_{z \in \mathcal{I}(\mathcal{C})} \left[\sup_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma) - \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma) \right]. \tag{25}$$

Proof. For fixed z, Theorem 1 gives the same value $c^{\star}_{\mathcal{I}}(z) = \inf_{\Sigma' \in \mathcal{F}_{\mathcal{I}}(z)} m(\Sigma')$ for every system in $\mathcal{F}_{\mathcal{I}}(z)$. Taking the supremum of $m(\Sigma) - c^{\star}_{\mathcal{I}}(z)$ over that class proves (24). Taking the supremum over all nonempty information classes gives (25). □

Theorem 2 shows that the relevant "size" of an equivalence class is not its geometric diameter in the full system space, but its width along the stability task m. A large unresolved set can have negligible information-induced conservatism if m varies little over it, whereas a narrow class can still be costly when it crosses a steep stability-relevant direction.

The zero-width case gives the exact condition under which the stability margin itself can be evaluated on the quotient space.

Theorem 3 (Exact Certificate on the Information Quotient)**.** *Let $q_{\mathcal{I}} : \mathcal{C} \to \mathcal{C}/\!\sim_{\mathcal{I}}$ denote the natural projection $q_{\mathcal{I}}(\Sigma) = [\Sigma]_{\mathcal{I}}$. The following statements are equivalent:*

1. *the information structure introduces no irreducible conservatism, i.e., $\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}) = 0$ for all $\Sigma \in \mathcal{C}$;*
2. *the ground-truth margin is constant on every information fiber, i.e.,*

$$\Sigma_1 \sim_{\mathcal{I}} \Sigma_2 \quad \Longrightarrow \quad m(\Sigma_1) = m(\Sigma_2); \tag{26}$$

3. *there exists a unique map $\bar{m}_{\mathcal{I}} : \mathcal{C}/\!\sim_{\mathcal{I}} \to \mathbb{R}$ such that*

$$m = \bar{m}_{\mathcal{I}} \circ q_{\mathcal{I}}. \tag{27}$$

Hence exact stability-margin evaluation from the compressed information is possible if and only if the stability task itself is well defined on the information-induced quotient space.

Proof. If $\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}) = 0$ for every Σ, then (12) gives $c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) = m(\Sigma)$. Therefore, any two systems in the same information fiber have the same value of m, which proves (26). Conversely, if m is constant on every fiber, then (8) gives $c^\star_{\mathcal{I}}(\mathcal{I}(\Sigma)) = m(\Sigma)$ and thus $\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}) = 0$. Finally, a function on $\mathcal{C}$ is constant on the equivalence classes of $\sim_{\mathcal{I}}$ if and only if it factors uniquely through the quotient map $q_{\mathcal{I}}$, which establishes the equivalence with (27). □

Equation (27) means that exact certificate is possible precisely when the ground-truth margin does not distinguish any two systems that the information structure has already declared indistinguishable. If this factorization fails, no refinement of the certificate map c alone can eliminate the resulting conservatism; the information structure itself must be refined.

The same quotient-space result immediately contains the binary stability problem as a special case.

Example 4 (Binary Stability as Quotient Separation)**.** *For binary stability certificate, define*

$$\begin{aligned} \mathcal{C}_{\mathrm{s}} &:= \{\Sigma \in \mathcal{C} : m(\Sigma) > 0\}, \\ \mathcal{C}_{\mathrm{u}} &:= \{\Sigma \in \mathcal{C} : m(\Sigma) \le 0\}. \end{aligned} \tag{28}$$

Unlike exact margin evaluation, a binary decision does not require m to be constant on each equivalence class. It only requires the information compression to preserve the stability boundary:

$$\boxed{q_{\mathcal{I}}(\mathcal{C}_{\mathrm{s}}) \cap q_{\mathcal{I}}(\mathcal{C}_{\mathrm{u}}) = \varnothing}. \tag{29}$$

Equivalently, every equivalence class must lie entirely on one side of $m = 0$. Hence binary stability is the natural sign-only consequence of Theorem 3, rather than a separate information concept.

Corollary 1 (Indistinguishable Stable/Unstable Pair)**.** *Suppose that there exist $\Sigma_{\mathrm{s}}, \Sigma_{\mathrm{u}} \in \mathcal{C}$ such that*

$$\mathcal{I}(\Sigma_{\mathrm{s}}) = \mathcal{I}(\Sigma_{\mathrm{u}}), \qquad m(\Sigma_{\mathrm{s}}) > 0, \qquad m(\Sigma_{\mathrm{u}}) \le 0. \tag{30}$$

Then no sound method using only $\mathcal{I}$ can certify Σ_{s} as stable.

Proof. Both systems belong to the same information fiber $\mathcal{F}_{\mathcal{I}}(\Sigma_{\mathrm{s}}) = \mathcal{F}_{\mathcal{I}}(\Sigma_{\mathrm{u}})$. Equation (8) gives $c^{\star}_{\mathcal{I}} \leq m(\Sigma_{\mathrm{u}}) \leq 0$ on that class. □

Corollary 1 is the basic information wall: the limitation is not that a particular distributed test fails, but that the available information does not distinguish the stable system from a physically admissible unstable one. It also converts the quotient-space separation condition above into a constructive impossibility test. Rather than characterizing the entire quotient space, it is sufficient to exhibit a single stable/unstable pair that collapses to the same equivalence class. Such a pair is therefore a witness that the current information structure is fundamentally insufficient, irrespective of the stability criterion used. This pairwise construction will be used in Section 5.

3.3 Information–Conservatism Frontier under Admissible Refinements

The preceding results evaluate the limitation created by a *fixed* information structure. To turn this benchmark into a reporting-interface-design problem, the admissible ways of enriching the report must first be specified. This restriction is essential: if arbitrary information structures were allowed, the single real-valued report $\mathcal{I}(\Sigma) = m(\Sigma)$ would make the information-induced conservatism identically zero, so neither the number of reported scalars nor the raw dimension of the report defines a meaningful information budget by itself.

Fix a baseline reporting interface with information structure $\mathcal{I}_0$ and a finite family of admissible information primitives

$$\Lambda = \{\mathcal{J}_1, \ldots, \mathcal{J}_p\}. \tag{31}$$

For a selected index set $A \subseteq [p] := \{1, \ldots, p\}$, define the refined information structure

$$\mathcal{I}_A := \{\mathcal{I}_0, \mathcal{J}_\ell : \ell \in A\}, \qquad \mathrm{cost}(A) := \sum_{\ell \in A} w_\ell, \quad w_\ell > 0, \tag{32}$$

where w_ℓ is the cost assigned to the ℓ-th reporting primitive. In practical systems, w_ℓ corresponds to the costs for adding monitoring equipments and corresponding communication links. For example, in power systems, it might be PMU or SCADA equipments. In the sense of information, the cost may represent one additional directional report, one certified cross-node relation, one model block, or another reporting-interface resource.

Definition 8 (Information–Conservatism Frontier)**.** *For an information budget $B \geq 0$, define the worst-case frontier*

$$\boxed{\Phi(B) := \min_{A \subseteq [p]:\, \mathrm{cost}(A) \leq B} \overline{\Delta}_{\mathrm{info}}(\mathcal{I}_A)}. \tag{33}$$

For a fixed physical system Σ, its point-wise counterpart is

$$\Phi(B; \Sigma) := \min_{A \subseteq [p]:\, \mathrm{cost}(A) \leq B} \Delta_{\mathrm{info}}(\Sigma; \mathcal{I}_A). \tag{34}$$

Conversely, for a prescribed tolerance $\varepsilon \geq 0$, define the minimum information cost

$$\boxed{\mathrm{cost}^{\star}(\varepsilon) := \min_{A \subseteq [p]} \left\{ \mathrm{cost}(A) : \overline{\Delta}_{\mathrm{info}}(\mathcal{I}_A) \leq \varepsilon \right\},} \tag{35}$$

with $\mathrm{cost}^{\star}(\varepsilon) = +\infty$ when the admissible refinement family cannot attain the requested tolerance.

By Theorem 2, the frontier has the equivalent partition-width form

$$\Phi(B) = \min_{A:\, \mathrm{cost}(A) \leq B} \sup_{z \in \mathcal{I}_A(\mathcal{C})} \left[\sup_{\Sigma \in \mathcal{F}_{\mathcal{I}_A}(z)} m(\Sigma) - \inf_{\Sigma \in \mathcal{F}_{\mathcal{I}_A}(z)} m(\Sigma) \right]. \tag{36}$$

Hence the information-design problem is geometrically a partition-design problem: under a prescribed reporting budget, refine the information fibers so that their largest width along the task m is as small as possible. The frontier is nonincreasing in B, and the two formulations obey

$$\Phi(B) \le \varepsilon \quad \Longleftrightarrow \quad \mathrm{cost}^\star(\varepsilon) \le B. \tag{37}$$

In particular, $\mathrm{cost}^\star(0)$ is the minimum admissible reporting-interface cost required for exact task sufficiency.

The minimum-information problem also admits a pairwise characterization that extends Corollary 1. For $\varepsilon \ge 0$, define the baseline task-relevant witness set

$$\begin{aligned}\mathcal{W}_\varepsilon := \big\{(\Sigma_a, \Sigma_b) \in \mathcal{C}^2 : \mathcal{I}_0(\Sigma_a) = \mathcal{I}_0(\Sigma_b),\\ |m(\Sigma_a) - m(\Sigma_b)| > \varepsilon\big\}.\end{aligned} \tag{38}$$

and, for each pair, the primitives capable of separating it,

$$D(\Sigma_a, \Sigma_b) := \{\ell \in [p] : \mathcal{J}_\ell(\Sigma_a) \ne \mathcal{J}_\ell(\Sigma_b)\}. \tag{39}$$

Theorem 4 (Task-Relevant Witness Cover)**.** *For any selected refinement $A \subseteq [p]$ and any $\varepsilon \ge 0$,*

$$\boxed{\overline{\Delta}_{\mathrm{info}}(\mathcal{I}_A) \le \varepsilon \quad \Longleftrightarrow \quad A \cap D(\Sigma_a, \Sigma_b) \ne \varnothing,\ \forall (\Sigma_a, \Sigma_b) \in \mathcal{W}_\varepsilon.} \tag{40}$$

Proof. Suppose first that a pair $(\Sigma_a, \Sigma_b) \in \mathcal{W}_\varepsilon$ is not separated by any primitive in A. Since the two systems already have the same baseline report, they also satisfy $\mathcal{I}_A(\Sigma_a) = \mathcal{I}_A(\Sigma_b)$ and therefore belong to the same $\mathcal{I}_A$-fiber. The width of that fiber along m is strictly larger than ε, so Theorem 2 gives $\overline{\Delta}_{\mathrm{info}}(\mathcal{I}_A) > \varepsilon$.

Conversely, suppose $\overline{\Delta}_{\mathrm{info}}(\mathcal{I}_A) > \varepsilon$. By (25), some $\mathcal{I}_A$-fiber has m-width greater than ε. Hence it contains two systems Σ_a, Σ_b satisfying $|m(\Sigma_a) - m(\Sigma_b)| > \varepsilon$. They have the same baseline report and are not distinguished by any selected primitive, so $(\Sigma_a, \Sigma_b) \in \mathcal{W}_\varepsilon$ and $A \cap D(\Sigma_a, \Sigma_b) = \varnothing$. □

Theorem 4 makes the frontier constructive: additional information has value only through the task-relevant pairs that it separates. For a finite admissible system set, (35) is exactly the weighted covering problem

$$\begin{aligned}\min_{x_1,\dots,x_p} \quad & \sum_{\ell=1}^{p} w_\ell x_\ell \\ \text{s.t.} \quad & \sum_{\ell \in D(\Sigma_a,\Sigma_b)} x_\ell \ge 1, \quad \forall (\Sigma_a, \Sigma_b) \in \mathcal{W}_\varepsilon, \\ & x_\ell \in \{0,1\}, \qquad\qquad \ell = 1, \dots, p.\end{aligned} \tag{41}$$

For a continuous system class, the same statement becomes a semi-infinite covering problem and suggests a constraint-generation interpretation: a candidate reporting interface is refined only when an adversarial indistinguishable pair with task separation greater than ε can still be found. For binary stability certification, $\mathcal{W}_\varepsilon$ is replaced by the stable/unstable witness pairs of Corollary 1; thus minimum sufficient information is precisely the minimum-cost refinement that separates every such pair.

4 Network-Node Feedback Interconnection Structure and a Unified View of Information-Induced Conservatism

Distributed stability methods exploit the modular structure of an interconnected system. A common physical representation separates local node dynamics from network interconnections:

$$\begin{aligned}\text{node: } & y = G \circ u,\ G = \mathrm{blkdiag}\big(G_1, \dots, G_n\big), \\ \text{network: } & u = -N \circ y,\end{aligned} \tag{42}$$

Here G_i is the dynamics of node i and N describes the network interconnection. This representation is widely used in practical system modeling such as power grids [3, 14]. Different port selections lead to different realizations, but the architectural principle is the same: local dynamics and network interactions are separated so that certification can be assembled modularly.

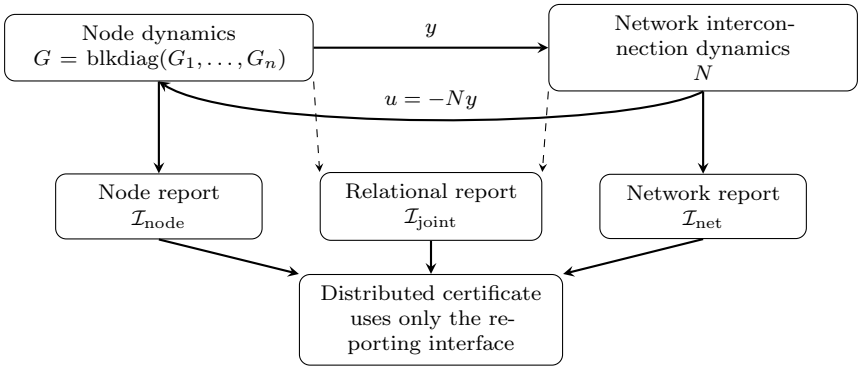


Figure 3: Physical decomposition and reporting-interface decomposition in distributed stability certification. The network–node model specifies how the physical system is assembled; the reporting interface induces the information structure that specifies what the certificate can distinguish. The three studies below correspond to compression of the network report, compression of the node report, and omission of relational information, respectively.

For the present paper, the crucial point is that the reporting interface induces an information structure of the form

$$\mathcal{I}(\Sigma) = \left\{\mathcal{I}_{\mathrm{node}},\ \mathcal{I}_{\mathrm{net}},\ \mathcal{I}_{\mathrm{joint}}\right\},$$

where the first two terms represent the node and network reports and $\mathcal{I}_{\mathrm{joint}}$ represents any retained relation between them or across nodes. A practical reporting interface may compress either side and may omit $\mathcal{I}_{\mathrm{joint}}$ entirely. Distributed certification is therefore not only a problem of exploiting a physical decomposition; it is also a problem of designing the reporting interface that determines what the certificate is allowed to observe.

The physical decomposition does not, by itself, determine what the certificate knows. The three application studies therefore vary the *reporting interface*, not the underlying idea of modular interconnection. Table 1 places the three sources of information-induced conservatism side by side and makes explicit the relation that each reporting interface fails to preserve.

The three cases expose a common structural message: stability can depend on relations among quantities reported by all interfaces. Network-side compression loses the placement of network directions relative to heterogeneous nodes; node-side compression loses local eigendirections relative to the critical interconnection direction; cross-node compression loses relations that do not belong to any single local block. In every case, the information structure places systems with potentially different stability margins in the same information fiber. Consequently, the stability value of information is governed by how strongly the task varies along the unresolved relation, rather than by the number of scalars added to or removed from the report.

This interpretation leads to a concrete diagnostic workflow. *Step 1:* specify the stability task and its margin. *Step 2:* write the information structure induced by the reporting interface, including what joint relations are unavailable. *Step 3:* characterize the adverse variation inside the resulting information fiber, either by computing $c^\star_{\mathcal{I}}$ or by constructing an indistinguishable witness pair. *Step 4:* compare the two gaps in (14). Large method-induced conservatism Δ_{method} calls for a better criterion or optimization; large information-induced conservatism Δ_{info} calls for a richer reporting interface; if both terms are small, the existing reporting interface is already close to the task-relevant limit. The following sections apply this workflow to the three reporting interfaces in Table 1.

Table 1: Three representative sources of information-induced conservatism in a modular network–node architecture. The common issue is not simply that quantities are omitted, but that stability-relevant relations become unresolved.

Source of conservatism	What remains available	Stability-relevant relation left unresolved	Studied in
Network-side	Complete node models and the network Laplacian eigenvalues	Alignment between network eigendirections and heterogeneous node characteristics	Section 5
Node-side	Complete network certificate information and node scalar or eigenvalue-only passivity reports	Alignment between node passivity eigendirections and the critical interconnection direction	Section 6
Cross-node	Exact local matrix blocks assembled from independently certified nodes	Cross-node quadratic relations that cannot be reconstructed from the local blocks alone	Section 7

5 Network-Side Information-Induced Conservatism under Spectral Compression

The preceding results characterize the best stability certificate permitted by a given information structure. We now examine a common source of information-induced conservatism on the network side of distributed assessment: replacing the network matrix by its eigenvalue spectrum. To isolate this compression effect, the node side is assumed to be known *completely*. The question is whether network eigenvalues alone, without the associated eigenvectors or their alignment with heterogeneous nodes, are sufficient for exact stability certificate.

5.1 General Heterogeneous Systems

Consider the linearized system dynamics $y = G(s)u, \qquad u = -bLy$, where $L = L^T \succeq 0$ is the Laplacian of a connected network, $b > 0$ is a strength scaling, and the complete node dynamics are

$$G(s) = g_0(s)K, \qquad K = \mathrm{diag}(k_1, \ldots, k_n) \succ 0. \tag{43}$$

Although (43) is a restricted subclass of heterogeneous nodes, an impossibility result on this subclass is already sufficient to disprove universal sufficiency of network-spectrum information for heterogeneous systems.

Since KL is similar to $K^{1/2}LK^{1/2}$, the closed-loop characteristic determinant factorizes exactly as

$$\det[I + bG(s)L] = \det[I + bg_0(s)K^{1/2}LK^{1/2}] = \prod_{j=1}^{n} \left[1 + b\mu_j g_0(s)\right], \qquad \mu_j \in \mathrm{spec}(K^{1/2}LK^{1/2}). \tag{44}$$

Thus, once the nodes are heterogeneous, the stability-relevant network quantity is generally the matched spectrum $\mathrm{spec}(K^{1/2}LK^{1/2})$, rather than $\mathrm{spec}(L)$ alone. Actually, we can derive a more impressive result as follows.

Proposition 3 (Network Spectrum Is Not Universally Sufficient under Heterogeneity)**.** *Fix any $n \geq 4$ and a stable, real-rational, strictly proper common node shape $g_0(s) = n(s)/d(s)$. Suppose that the scalar modal family*

$$\chi_\mu(s) := d(s) + \mu n(s), \qquad \mu \geq 0, \tag{45}$$

has a finite first positive-gain loss-of-Hurwitz boundary

$$0 < \mu_c := \inf\{\mu > 0 : \chi_\mu(s) \text{ is not Hurwitz}\} < \infty. \tag{46}$$

Then there exist a heterogeneous gain matrix $K \succ 0$ and two connected n-node Laplacians L_A, L_B such that

$$\operatorname{spec}(L_A) = \operatorname{spec}(L_B),$$

but one closed-loop interconnection is asymptotically stable while the other is not Hurwitz. Consequently, the information structure

$$\mathcal{I}^L_{\text{spec}}(\Sigma) := \{G_1(s), \dots, G_n(s), \operatorname{spec}(L)\} \tag{47}$$

is not universally sufficient for exact stability evaluation on this heterogeneous system class. A sufficient condition for the finite-boundary assumption above is that g_0 has relative degree at least three.

Proof. In the proof, we consider the cases with star topology demonstrated in Section 1. Because d is Hurwitz, χ_μ remains Hurwitz for all sufficiently small $\mu \geq 0$, so $\mu_c > 0$. The theorem assumes that a finite loss-of-Hurwitz boundary exists. The commonly encountered relative-degree condition $r := \deg d - \deg n \geq 3$ guarantees this finiteness: the root locus of $d(s) + \mu n(s)$ has r branches tending to infinity, and, for either sign of the high-frequency gain, at least one asymptote enters the open right-half plane when $r \geq 3$. Hence χ_μ is non-Hurwitz for some finite positive μ.

It remains to construct two networks with distinct stability and the same spectrum. Let

$$K = \operatorname{diag}(\kappa, 1, \dots, 1), \qquad \kappa > 1.$$

Consider an unweighted n-node star. In Network A, node 1, carrying the gain κ, is the center. In Network B, node 2 is the center, so the same labeled heterogeneous node 1 becomes a leaf. The two Laplacians have exactly the same complete spectrum,

$$\operatorname{spec}(L_A) = \operatorname{spec}(L_B) = \{0, 1, \dots, 1, n\}. \tag{48}$$

The node models remain fixed at their labeled nodes and are therefore identical in the two systems.

For Network A,

$$\lambda_{L,A} := \lambda_{\max}(K^{1/2} L_A K^{1/2}) = \kappa(n-1) + 1. \tag{49}$$

For Network B, the $n-3$ directions among the ordinary leaves have eigenvalue 1, while the remaining nonzero matched eigenvalues are the roots of $p(\lambda) = \lambda^2 - (n+\kappa)\lambda + \kappa(n-1) + 1$. Hence

$$\lambda_{L,B} = \frac{n + \kappa + \sqrt{(n+\kappa)^2 - 4[\kappa(n-1)+1]}}{2}. \tag{50}$$

Substituting $\lambda_{L,A}$ gives

$$p(\lambda_{L,A}) = (\kappa - 1)(n-2)[\kappa(n-1) + 1] > 0. \tag{51}$$

Moreover, for $n \geq 4$ and $\kappa > 1$, $\lambda_{L,A} > (n+\kappa)/2$, so $\lambda_{L,A}$ lies to the right of the vertex of p. Therefore

$$\lambda_{L,A} > \lambda_{L,B}. \tag{52}$$

Choose

$$b = \frac{\mu_c}{\lambda_{L,A}}.$$

For Network B, every matched eigenvalue is nonnegative and satisfies

$$b\mu_j \leq b\lambda_{L,B} = \mu_c \frac{\lambda_{L,B}}{\lambda_{L,A}} < \mu_c. \tag{53}$$

By the definition of the first boundary, every scalar factor in (44) is Hurwitz, and Network B is asymptotically stable. In Network A, the largest matched mode satisfies $b\lambda_{L,A} = \mu_c$. Since the set of Hurwitz polynomials is open and the coefficient map is continuous, the boundary polynomial χ_{μ_c} is non-Hurwitz. Hence Network A is not Hurwitz. This produces an indistinguishable stable/unstable pair under $\mathcal{I}^L_{\text{spec}}$ and proves the result. □

The theorem should be read as a statement about *universal information sufficiency*, not as a claim that every heterogeneous node model produces an isospectral stability mismatch. A single admissible heterogeneous subclass containing such a pair is enough to show that the report $\{G_i, \mathrm{spec}(L)\}$ cannot support an exact certificate that is valid for the whole class. *The missing quantity is the alignment between network modes and heterogeneous node characteristics.* This distinction disappears in the homogeneous limit $K = kI$, because then $K^{1/2}LK^{1/2} = kL$ and the matched spectrum is determined completely by $\mathrm{spec}(L)$. Thus the limitation is not caused by spectral analysis itself, but by compressing away a node–network relation that becomes stability-relevant under heterogeneity.

5.2 Quantifying the Conservatism of Spectral Compression

The preceding theorem proves that the network spectrum can be insufficient. The information framework also allows the resulting conservatism to be quantified. For the common-shape class in (43), consider the margin

$$m(\Sigma) = \mu_c - b\lambda_{\max}(K^{1/2}LK^{1/2}), \tag{54}$$

where μ_c is defined in (46).

If the full labeled network matrix L is available, this margin is known exactly. If only the spectral report (47) is available, the node matrix K remains fixed while L can vary over all admissible isospectral networks in the same information fiber. Therefore, Theorem 1 gives

$$c^{\star}_{\mathcal{I}^L_{\mathrm{spec}}}(\Sigma) = \mu_c - b \sup_{\Sigma' \in \mathcal{F}_{\mathcal{I}^L_{\mathrm{spec}}}(\Sigma)} \lambda_{\max}\Big(K^{1/2}L'K^{1/2}\Big). \tag{55}$$

Consequently, the exact conservatism introduced by compressing the network matrix into its spectrum is

$$\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}^L_{\mathrm{spec}}) = b\Big[\sup_{\Sigma' \in \mathcal{F}_{\mathcal{I}^L_{\mathrm{spec}}}(\Sigma)} \lambda_{\max}(K^{1/2}L'K^{1/2}) - \lambda_{\max}(K^{1/2}LK^{1/2})\Big]. \tag{56}$$

Every node permutation P^TLP, while keeping the node matrix K fixed at the labeled nodes, is a physically admissible Laplacian with the same spectrum. Hence

$$\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}^L_{\mathrm{spec}}) \geq b\Big[\max_{P \text{ permutation}} \lambda_{\max}(K^{1/2}P^TLPK^{1/2}) - \lambda_{\max}(K^{1/2}LK^{1/2})\Big]. \tag{57}$$

The lower bound isolates the conservatism caused purely by the unresolved alignment of known nodes relative to a network with the same spectrum.

Example 5. *The general result above can be seen in a four-node construction for which all quantities are available in closed form. Fix the labeled node dynamics as* $G_i(s) = k_i/(s+1)^3$ *with* $(k_1, k_2, k_3, k_4) = (\kappa, \kappa, 1, 1)$ *and place them into the two networks in Fig. 4. Network A places the two high-gain nodes on adjacent nodes of the ring, whereas Network B places them on opposite nodes.*

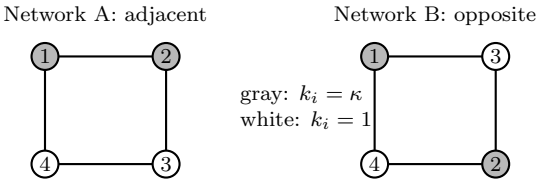


Figure 4: Four-node pair with identical labeled node dynamics and identical complete Laplacian spectrum. Only the alignment between node heterogeneity and network modes changes.

With unit line weights, the two Laplacians are

$$L_A = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix}, \; L_B = \begin{bmatrix} 2 & 0 & -1 & -1 \\ 0 & 2 & -1 & -1 \\ -1 & -1 & 2 & 0 \\ -1 & -1 & 0 & 2 \end{bmatrix}. \tag{58}$$

They are similar and satisfy $\operatorname{spec}(L_A) = \operatorname{spec}(L_B) = \{0, 2, 2, 4\}$. *Nevertheless, for* $K = \operatorname{diag}(\kappa, \kappa, 1, 1)$, *the largest matched eigenvalues are*

$$\lambda_{L,A} = \frac{3(\kappa+1) + \sqrt{9\kappa^2 - 14\kappa + 9}}{2}, \qquad \lambda_{L,B} = 2(\kappa+1). \tag{59}$$

Their difference is strict for every $\kappa > 1$ *because* $9\kappa^2 - 14\kappa + 9 - (\kappa+1)^2 = 8(\kappa-1)^2 > 0$. *Take* $\kappa = 2.75$ *and* $b = 1$. *Then* $\lambda_{L,A} \approx 8.72994$ *and* $\lambda_{L,B} = 7.5$. *For* $g_0(s) = 1/(s+1)^3$, *the Routh–Hurwitz criterion gives the first positive boundary* $\mu_c = 8$. *Hence Network A is unstable while Network B is asymptotically stable.*

To quantify the resulting information-induced conservatism, take Network B as the true system. Its true first-boundary margin is $m(\Sigma_B) = 8 - \lambda_{L,B} = 0.5$. *The same spectral fiber contains Network A, for which* $\lambda_{L,A} \approx 8.72994$. *Thus*

$$\Delta_{\text{info}}(\Sigma_B; \mathcal{I}^L_{\text{spec}}) \geq \lambda_{L,A} - \lambda_{L,B} \approx 1.22994, \tag{60}$$

and consequently $c^\star_{\mathcal{I}^L_{\text{spec}}}(\Sigma_B) \leq -0.72994$. *Hence the spectrum-only reporting interface loses more stability margin than the true system possesses: although Network B is stable with margin* 0.5, *no sound certificate based only on the common spectrum can certify it.*

6 Node-Side Information-Induced Conservatism under Spectral Compression

The network-side result above shows that spectral compression can discard stability-relevant eigenvector alignment. We next examine the complementary node-side information-induced conservatism caused by compressing matrix passivity indices into scalar reports. This section uses passivity-based certificates for demonstration, but the underlying mechanism is not specific to passivity.

Consider the stacked passivity condition associated with Example 1,

$$\Pi - \Pi_g \succ 0, \qquad \Pi = \operatorname{blkdiag}(\Pi_1, \ldots, \Pi_n), \tag{61}$$

where $\Pi_i \in \mathbb{S}^2$ and $\Pi_g \in \mathbb{S}^{2n}$ denote the node and network passivity indices, respectively. For this comparison, use the passivity-residual margin

$$m(\Sigma) = \lambda_{\min}(\Pi - \Pi_g) \tag{62}$$

as the task margin. The full matrix report therefore provides the benchmark against the information-induced conservatism caused by scalarization.

6.1 Passivity-Based Node Reports with Scalar, Spectral, and Matrix Values

For each node, recall the scalar report $\pi_i = \lambda_{\min}(\Pi_i)$. Since $\Pi_i \in \mathbb{S}^2$, it can always be written as

$$\begin{aligned} &\Pi_i = \pi_i I_2 + r_i s_i s_i^T, \qquad \|s_i\|_2 = 1, \\ &r_i = \lambda_{\max}(\Pi_i) - \pi_i \geq 0. \end{aligned} \tag{63}$$

The full eigenvalue spectrum is $\{\pi_i, \pi_i + r_i\}$. Hence, the scalar reporting interface retains only π_i, whereas a complete eigenvalue-spectrum report additionally reveals $\pi_i + r_i$ but still does not reveal the direction s_i. Define

$$\Pi_{\mathrm{sc}} := \mathrm{blkdiag}(\pi_1 I_2, \ldots, \pi_n I_2). \tag{64}$$

To avoid confusing this node-side spectrum with the network spectrum used in Section 5, denote by $\mathcal{I}^{\Pi}_{\mathrm{spec}}$ the refinement of $\mathcal{I}^{\Pi}_{\mathrm{sc}}$ that reports the complete eigenvalue spectrum of every Π_i, but not their eigenvectors.

Although $\mathcal{I}^{\Pi}_{\mathrm{spec}} \succeq \mathcal{I}^{\Pi}_{\mathrm{sc}}$, the two information structures have exactly the same best sound certificate:

$$c^{\star}_{\mathcal{I}^{\Pi}_{\mathrm{spec}}} = c^{\star}_{\mathcal{I}^{\Pi}_{\mathrm{sc}}} = \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g). \tag{65}$$

To see this, every matrix report Π' compatible with either report satisfies $\Pi' \succeq \Pi_{\mathrm{sc}}$, and therefore

$$\lambda_{\min}(\Pi' - \Pi_g) \geq \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g). \tag{66}$$

For the full spectral report, let v be a unit eigenvector associated with $\lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g)$ and partition it according to the node ports as $v = [v_1^T, \ldots, v_n^T]^T$, with $v_i \in \mathbb{R}^2$. The unreported eigenvector s_i in (63) can always be chosen orthogonal to v_i. Then

$$v^T(\Pi' - \Pi_{\mathrm{sc}})v = \sum_{i=1}^{n} r_i (s_i^T v_i)^2 = 0. \tag{67}$$

Thus $v^T(\Pi' - \Pi_g)v = \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g)$. By the Rayleigh–Ritz characterization, $\lambda_{\min}(\Pi' - \Pi_g) \leq v^T(\Pi' - \Pi_g)v$; together with the preceding lower bound, equality follows. Hence (65) holds even when every r_i is known exactly. Equivalently,

$$\Delta c^*(\mathcal{I}^{\Pi}_{\mathrm{spec}} \mid \mathcal{I}^{\Pi}_{\mathrm{sc}}; \Sigma) = 0. \tag{68}$$

This result makes the distinction between the *amount* and the *stability value* of information explicit. Reporting the full spectrum may add many exact numbers, yet it does not reduce the worst-case conservatism unless the associated eigenvectors are also resolved.

6.2 Stability Value of Matrix-Valued Node Information

The full matrix report removes the rotational ambiguity retained by the scalar and spectral reports. Therefore,

$$c^{\star}_{\mathcal{I}^{\Pi}_{\mathrm{mat}}} = \lambda_{\min}(\Pi - \Pi_g) = m(\Sigma), \tag{69}$$

and the exact stability value of upgrading from scalar to matrix passivity information is

$$\Delta c^*(\mathcal{I}^{\Pi}_{\mathrm{mat}} \mid \mathcal{I}^{\Pi}_{\mathrm{sc}}; \Sigma) = \lambda_{\min}(\Pi - \Pi_g) - \lambda_{\min}(\Pi_{\mathrm{sc}} - \Pi_g) \geq 0. \tag{70}$$

This is exactly the information-induced conservatism introduced by scalarizing the matrix passivity reports into scalar ones.

Let v_{sc} and v_{mat} be unit eigenvectors associated with the smallest eigenvalues of $\Pi_{\mathrm{sc}} - \Pi_g$ and $\Pi - \Pi_g$, respectively, and denote their i-th node blocks by v_i^{sc} and v_i^{mat}. The same variational eigenvalue reasoning gives

$$\sum_{i=1}^{n} r_i (s_i^T v_i^{\mathrm{mat}})^2 \leq \Delta c^*(\mathcal{I}^{\Pi}_{\mathrm{mat}} \mid \mathcal{I}^{\Pi}_{\mathrm{sc}}; \Sigma) \leq \sum_{i=1}^{n} r_i (s_i^T v_i^{\mathrm{sc}})^2. \tag{71}$$

Thus, the value of matrix passivity information is determined not by the additional eigenvalue surplus r_i alone, but by its alignment with the critical stability eigenvector.

6.3 Exact Frontier for Partial Directional Node Information

The scalar–spectral–matrix comparison above gives only the two endpoints of a richer design question: if the complete eigenvalue spectrum of every Π_i is available, but the reporting interface can reveal the eigendirections of only a subset of nodes, which directions should be reported and how much conservatism remains? This produces an exact instance of the general frontier in Section 3.3.

Using (63), define the embedded directional surplus

$$Q_i := \mathrm{blkdiag}(0, \dots, 0, r_i s_i s_i^T, 0, \dots, 0) \in \mathbb{S}^{2n}, \tag{72}$$

where the only nonzero 2×2 block is the i-th one. Then

$$\Pi = \Pi_{\mathrm{sc}} + \sum_{i=1}^{n} Q_i. \tag{73}$$

For $A \subseteq [n]$, let $\mathcal{I}_A^{\Pi}$ report Π_g, all eigenvalue pairs $\{\pi_i, \pi_i + r_i\}$, and in addition the eigendirections s_i for $i \in A$. Thus $A = \varnothing$ is the spectral reporting interface $\mathcal{I}_{\mathrm{spec}}^{\Pi}$, whereas $A = [n]$ is the full matrix reporting interface. Define

$$M_A := \Pi_{\mathrm{sc}} - \Pi_g + \sum_{i \in A} Q_i. \tag{74}$$

Proposition 4 (Optimal Certificate with Partially Revealed Eigendirections)**.** *For every $A \subseteq [n]$, the best sound certificate permitted by $\mathcal{I}_A^{\Pi}$ is*

$$\boxed{c^{\star}_{\mathcal{I}_A^{\Pi}} = \lambda_{\min}(M_A).} \tag{75}$$

Consequently, at the true system Σ,

$$\Delta_{\mathrm{info}}(\Sigma; \mathcal{I}_A^{\Pi}) = \lambda_{\min}(\Pi - \Pi_g) - \lambda_{\min}(M_A). \tag{76}$$

Proof. Every matrix report compatible with $\mathcal{I}_A^{\Pi}$ has the form

$$M_A + \sum_{i \notin A} \widetilde{Q}_i, \qquad \widetilde{Q}_i \succeq 0, \tag{77}$$

where $\widetilde{Q}_i$ has the same rank-one eigenvalue surplus r_i in its i-th block but an unresolved unit direction $\widetilde{s}_i$. Hence

$$\lambda_{\min}\left(M_A + \sum_{i \notin A} \widetilde{Q}_i\right) \geq \lambda_{\min}(M_A). \tag{78}$$

Let v be a unit eigenvector of M_A associated with $\lambda_{\min}(M_A)$ and partition it as $v = [v_1^T, \dots, v_n^T]^T$. For every unresolved node $i \notin A$, choose an admissible $\widetilde{s}_i \perp v_i$; if $v_i = 0$, any unit direction may be used. Then $v^T \widetilde{Q}_i v = r_i (\widetilde{s}_i^T v_i)^2 = 0$ for all $i \notin A$. Therefore the Rayleigh quotient of the compatible completion in (77) along v equals $\lambda_{\min}(M_A)$, which proves that the lower bound is attainable. The gap formula follows from (62). □

With unit cost per revealed node direction, define the k-direction point-wise frontier by

$$\Phi_{\Pi}(k; \Sigma) := \min_{A \subseteq [n]:\, |A| \leq k} \Delta_{\mathrm{info}}(\Sigma; \mathcal{I}_A^{\Pi}). \tag{79}$$

Proposition 4 yields the exact expression

$$\boxed{\Phi_{\Pi}(k; \Sigma) = m(\Sigma) - \max_{A \subseteq [n]:\, |A| \leq k} \lambda_{\min}\left(\Pi_{\mathrm{sc}} - \Pi_g + \sum_{i \in A} Q_i\right).} \tag{80}$$

It satisfies

$$\Phi_\Pi(0;\Sigma) = \Delta_{\text{info}}(\Sigma;\mathcal{I}^\Pi_{\text{spec}}) = \Delta_{\text{info}}(\Sigma;\mathcal{I}^\Pi_{\text{sc}}), \qquad \Phi_\Pi(n;\Sigma) = 0, \tag{81}$$

and is nonincreasing in k. Thus the previous result that spectral information has zero value relative to the scalar reporting interface becomes the left endpoint of a complete directional-information frontier rather than an isolated comparison.

The best certified margin at budget k can be computed as the mixed-integer semidefinite program

$$\begin{aligned} \max_{x,t} \quad & t \\ \text{s.t.} \quad & \Pi_{\text{sc}} - \Pi_g + \sum_{i=1}^n x_i Q_i \succeq t I_{2n}, \\ & \sum_{i=1}^n x_i \le k, \qquad x_i \in \{0,1\}. \end{aligned} \tag{82}$$

Accordingly, when the full matrix reporting interface certifies a positive margin, the minimum number of node directions required to retain a positive certificate is

$$\boxed{k^\star_{\text{cert}} := \min_{A\subseteq[n]} \left\{ |A| : \lambda_{\min}\left(\Pi_{\text{sc}} - \Pi_g + \sum_{i\in A} Q_i\right) > 0 \right\}.} \tag{83}$$

This quantity gives a direct reporting-interface-design interpretation: the transition from scalar to full matrix passivity need not be all-or-nothing; only the node directions that move the certificate across the required margin need to be exposed.

6.4 Node–Network Direction Matching as Joint Information

The network- and node-side sources of conservatism above are two views of the same missing information. On the node side, the scalar passivity report replaces Π_i by

$$\Pi_{i,\text{sc}} = \pi_i I_2, \tag{84}$$

and discards the directional surplus

$$\Pi_i - \Pi_{i,\text{sc}} = r_i s_i s_i^T. \tag{85}$$

The conservatism introduced by scalar passivity can therefore be expressed through the eigenvalue perturbation between the scalar and matrix certificates. Let v_{sc} be the unit eigenvector associated with the smallest eigenvalue of $\Pi_{\text{sc}} - \Pi_g$, and denote its i-th node block by v_i^{sc}. By first-order eigenvalue perturbation,

$$\Delta c^*(\mathcal{I}^\Pi_{\text{mat}} \mid \mathcal{I}^\Pi_{\text{sc}};\Sigma) = \lambda_{\min}(\Pi - \Pi_g) - \lambda_{\min}(\Pi_{\text{sc}} - \Pi_g) = \sum_{i=1}^n r_i (s_i^T v_i^{\text{sc}})^2 + O(\|\Pi - \Pi_{\text{sc}}\|^2). \tag{86}$$

Therefore, the conservatism caused by scalarization is not determined by the discarded eigenvalue surplus r_i alone. It depends on how much of this surplus is aligned with the critical eigenvector of the scalar certificate.

For each node, the relevant directional information is

$$(s_i^T v_i)^2 = \|v_i\|^2 \cos^2\theta_i, \tag{87}$$

where θ_i denotes the relative angle between the passivity eigenvector s_i and the critical network/node-interconnection eigenvector v_i. If v_i is known but s_i remains unresolved in the information fiber, an

admissible $s_i \perp v_i$ makes (87) zero. Conversely, if s_i is known but the critical direction remains unresolved, an admissible $v_i \perp s_i$ produces the same worst case. Hence one-sided directional information has no stability value unless the relative alignment between the two sides is resolved.

This gives a common interpretation of the two preceding applications. In Section 5, $\mathrm{spec}(L)$ retains modal strengths but removes the network eigenvectors relative to heterogeneous node coordinates; the matched matrix $K^{1/2}LK^{1/2}$ restores that relation. On the node side, a scalar or eigenvalue-only passivity report retains local strengths but removes the passivity eigenvectors relative to the critical interconnection direction. The task-relevant object in both cases is therefore not an eigenvalue or eigenvector in isolation, but their *relative alignment*.

7 Cross-Node Information-Induced Conservatism under Block-Diagonal Compression

The preceding section studies the conservatism caused by compressing each node's coupling information into a scalar report, i.e., by neglecting *intra-node* directional information. We now move outward and ask how much additional conservatism is introduced by neglecting *cross-node* information. In particular, even if every local matrix Π_i is known exactly, the standard distributed passivity structure still restricts the node-side matrix to $\Pi = \mathrm{blkdiag}(\Pi_1, \ldots, \Pi_n)$. This section quantifies the stability value of the cross-node information discarded by this block-diagonal reporting interface.

An important distinction is needed first. Let $\mathcal{I}$ denote the information structure under which the node side provides strictly local information. Under independent local certificate, the local facts supporting Π_i imply the aggregate matrix Π and nothing stronger across nodes. Thus the block-diagonal structure is not an additional approximation once $\mathcal{I}$ has already been fixed: it is the mathematical consequence of the local reporting interface itself. In this sense, the classical distributed passivity composition is already optimal under the predetermined $\mathcal{I}$. The question below is instead the value of refining $\mathcal{I}$ by additional certified cross-node quadratic information $\mathcal{J}$, i.e., the price of distributization measured by Definition 7.

Under the local information structure, the best sound passivity certificate is

$$c^\star_{\mathcal{I}}(\Sigma) = \lambda_{\min}(\Pi - \Pi_g). \tag{88}$$

To form an information-limited benchmark for adding joint information, keep all local diagonal blocks Π_i fixed and allow the refined reporting interface to certify a full symmetric joint matrix $\widetilde{\Pi}$ satisfying $\widetilde{\Pi}_{ii} = \Pi_i$. The best margin available under this idealized joint refinement is then

$$c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) = \sup_{\substack{\widetilde{\Pi}=\widetilde{\Pi}^T \\ \widetilde{\Pi}_{ii}=\Pi_i}} \lambda_{\min}(\widetilde{\Pi} - \Pi_g) = \min_i \lambda_{\min}(\Pi_i - (\Pi_g)_{ii})\,. \tag{89}$$

Indeed, the i-th principal block of $\widetilde{\Pi} - \Pi_g$ is always $\Pi_i - (\Pi_g)_{ii}$, so its minimum eigenvalue cannot exceed the right-hand side of (89). Equality is attained by choosing $\widetilde{\Pi}_{ij} = (\Pi_g)_{ij}$ for $i \neq j$, which cancels all cross-node blocks of the network term and leaves $\widetilde{\Pi} - \Pi_g$ block diagonal.

Therefore, the exact stability value of the cross-node information is

$$\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) = c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) - c^\star_{\mathcal{I}}(\Sigma) = \min_i \lambda_{\min}(\Pi_i - (\Pi_g)_{ii}) - \lambda_{\min}(\Pi - \Pi_g) \geq 0. \tag{90}$$

Hence, the price of decentralization is not a separate algebraic quantity: in the terminology of this paper, it is precisely the stability value of the cross-node relational information excluded by the local reporting interface. Equation (90) is the maximum certificate margin that can be recovered under the joint-information refinement above.

For compactness, define the block-diagonal and off-block parts of the network index by $\Pi_{g,\mathrm{bd}} := \mathrm{blkdiag}((\Pi_g)_{11}, \ldots, (\Pi_g)_{nn})$ and $\Pi_{g,\mathrm{off}} := \Pi_g - \Pi_{g,\mathrm{bd}}$. The role of network cross-node coupling is explicit.

By the Weyl inequality,

$$c^\star_{\mathcal{I}}(\Sigma) \geq c^\star_{\{\mathcal{I},\mathcal{J}\}}(\Sigma) - \|\Pi_{g,\mathrm{off}}\|_2. \tag{91}$$

Equivalently,

$$\boxed{0 \leq \Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) \leq \|\Pi_{g,\mathrm{off}}\|_2}. \tag{92}$$

Thus the off-block part of the network passivity matrix gives a global upper bound on the stability value of cross-node information. If the network has no cross-node coupling in these quadratic coordinates, then $\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0$ and the distributed block-diagonal restriction is lossless.

More generally, cross-node coupling can be nonzero while its stability value is still zero. From (90),

$$\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0 \tag{93}$$

if and only if

$$\Pi - \Pi_g \succeq \left[\min_i \lambda_{\min}(\Pi_i - (\Pi_g)_{ii})\right] I_{2n}. \tag{94}$$

This condition has a direct directional interpretation. Suppose node i attains the minimum in (89), and let x_i be a corresponding unit eigenvector of $\Pi_i - (\Pi_g)_{ii}$. Let x denote the full vector whose only nonzero block is x_i at node i. Whenever $\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0$, (94) implies

$$\Pi_{g,\mathrm{off}} x = 0. \tag{95}$$

Hence nonzero cross-node coupling does not by itself imply a positive stability value. If the cross-node network coupling does not act on the locally weakest directions and does not create a more adverse direction elsewhere, the local information structure already attains the joint optimum, so that $\Delta c^*(\mathcal{J} \mid \mathcal{I}; \Sigma) = 0$. The conservatism of distributization is therefore governed not only by the magnitude of the neglected cross-node coupling, but by whether that relational information acts along the critical local passivity directions.

8 Computable Local Approximation of Information Frontier

8.1 Local Unresolved Stability Information

The preceding sections define the exact information-limited benchmark through an optimization over the full fiber $\mathcal{F}_{\mathcal{I}}(z)$. That definition is conceptually sharp, but the corresponding optimization may be expensive in a large system. This section provides the computational bridge: when the unresolved part of an information fiber is locally parameterized, the information-induced conservatism can be approximated directly from the sensitivity of the stability task.

Fix an information structure $\mathcal{I}$ and consider the continuous ambiguity that remains inside one of its information fibers. Let $\vartheta \in \mathbb{R}^n$ collect the continuous quantities not exactly resolved by the current information structure. In the applications above, such coordinates may parameterize unresolved network–node alignment, rotations of local passivity eigendirections, or missing cross-node quadratic terms. The radius $\rho_{\mathcal{I}}$ below measures the local resolution of these hidden coordinates. Locally, write the fiber as

$$\vartheta + \mathcal{U}_{\mathcal{I}}, \qquad 0 \in \mathcal{U}_{\mathcal{I}}, \tag{96}$$

and define its radius

$$\rho_{\mathcal{I}} := \sup_{\delta \in \mathcal{U}_{\mathcal{I}}} \|\delta\|_2. \tag{97}$$

Definition 9 (Unresolved Stability Information)**.** *If $m(\vartheta)$ is differentiable at the nominal point, define*

$$\mathcal{S}_{\mathcal{I}}(\vartheta; m) = \sup_{\delta \in \mathcal{U}_{\mathcal{I}}} -\delta^T \nabla m(\vartheta). \tag{98}$$

When the task and nominal point are clear, it is abbreviated as $\mathcal{S}_{\mathcal{I}}$. The quantity $\mathcal{S}_{\mathcal{I}}$ has the same unit as m. It is the largest first-order stability-margin drop hidden by variations that the current information cannot distinguish.

Example 6. *A box-like fiber $|\delta_i| \le r_i$ yields*

$$\mathcal{S}_{\mathcal{I}} = \sum_{i=1}^{n} r_i \left| \frac{\partial m}{\partial \vartheta_i} \right|. \tag{99}$$

Thus raw uncertainty widths become comparable only after they are weighted by their relevance to the stability boundary.

Theorem 5 (Local Information–Conservatism Law)**.** *Assume that m is twice continuously differentiable on $\vartheta + \mathcal{U}_{\mathcal{I}}$ and that*

$$\|\nabla^2 m(\vartheta + \delta)\|_2 \le L_m, \qquad \forall \delta \in \mathcal{U}_{\mathcal{I}}. \tag{100}$$

Then

$$\mathcal{S}_{\mathcal{I}} - \frac{L_m}{2}\rho_{\mathcal{I}}^2 \le \Delta_{\text{info}}(\vartheta; \mathcal{I}) \le \mathcal{S}_{\mathcal{I}} + \frac{L_m}{2}\rho_{\mathcal{I}}^2. \tag{101}$$

In particular, as the information resolution increases such that $\rho_{\mathcal{I}} \to 0$,

$$\boxed{\Delta_{\text{info}} = \mathcal{S}_{\mathcal{I}} + O(\rho_{\mathcal{I}}^2)}. \tag{102}$$

Proof. Let $g = \nabla m(\vartheta)$. Taylor's theorem and (100) give, for every $\delta \in \mathcal{U}_{\mathcal{I}}$,

$$m(\vartheta) + g^T\delta - \frac{L_m}{2}\|\delta\|_2^2 \le m(\vartheta + \delta) \le m(\vartheta) + g^T\delta + \frac{L_m}{2}\|\delta\|_2^2. \tag{103}$$

Taking the infimum over δ and using

$$\inf_{\delta \in \mathcal{U}_{\mathcal{I}}} g^T\delta = -\mathcal{S}_{\mathcal{I}} \tag{104}$$

yields (101) after substitution into (23). □

Equivalently, the best certificate itself satisfies

$$m(\vartheta) - \mathcal{S}_{\mathcal{I}} - \frac{L_m}{2}\rho_{\mathcal{I}}^2 \le c_{\mathcal{I}}^\star \le m(\vartheta) - \mathcal{S}_{\mathcal{I}} + \frac{L_m}{2}\rho_{\mathcal{I}}^2. \tag{105}$$

Define the two endpoints as

$$\underline{c}_{\mathcal{I}} := m(\vartheta) - \mathcal{S}_{\mathcal{I}} - \frac{L_m}{2}\rho_{\mathcal{I}}^2,\ \bar{c}_{\mathcal{I}} := m(\vartheta) - \mathcal{S}_{\mathcal{I}} + \frac{L_m}{2}\rho_{\mathcal{I}}^2. \tag{106}$$

They provide an operational three-way decision. If $\underline{c}_{\mathcal{I}} > 0$, the current information is already sufficient to establish a positive sound margin. *If $\bar{c}_{\mathcal{I}} \le 0$, even the best sound certificate $c_{\mathcal{I}}^\star$ cannot be positive, so no alternative algorithm using the same information can certify stability.* If the interval $[\underline{c}_{\mathcal{I}}, \bar{c}_{\mathcal{I}}]$ crosses zero, the local bound is inconclusive and one should refine the information or tighten the local bound.

This relation turns the abstract benchmark into an operational local test. It estimates how much margin is hidden by the unresolved coordinates and therefore supports the same diagnosis developed above: whether the present reporting interface is already adequate for the task or whether resolving a particular stability-sensitive direction can materially improve certification.

8.2 Local Budgeted Information Allocation

The local law also gives an explicit approximation of the budgeted frontier. Consider the box-like ambiguity in Example 6 with $|\delta_i| \le r_i$, and suppose that one admissible information primitive resolves one hidden coordinate exactly. For a selected coordinate set $A \subseteq [n]$, the refined local fiber is

$$\mathcal{U}_A = \{\delta : \delta_i = 0 \text{ for } i \in A, \quad |\delta_i| \le r_i \text{ for } i \notin A\}. \tag{107}$$

Define the first-order value score

$$a_i := r_i \left|\frac{\partial m}{\partial \vartheta_i}\right|, \qquad a_{(1)} \ge a_{(2)} \ge \cdots \ge a_{(n)}, \tag{108}$$

where $(\cdot)$ denotes the decreasing ordering.

Proposition 5 (First-Order k-Information Frontier)**.** *Let $\rho_0^2 := \sum_{i=1}^n r_i^2$ and assume the Hessian bound in Theorem 5 holds uniformly on the baseline box. Under a unit-cost budget that resolves at most k coordinates, define*

$$\Phi_{\mathrm{loc}}(k;\vartheta) := \min_{A\subseteq[n]:\,|A|\le k} \Delta_{\mathrm{info}}(\vartheta;\mathcal{I}_A). \tag{109}$$

Then

$$\boxed{\Phi_{\mathrm{loc}}(k;\vartheta) = \sum_{j=k+1}^{n} a_{(j)} + O(\rho_0^2).} \tag{110}$$

Hence, to first order, the optimal information policy is to resolve the k coordinates with the largest scores $r_i|\partial m/\partial \vartheta_i|$.

Proof. For the refined fiber (107), Definition 9 gives

$$\mathcal{S}_{\mathcal{I}_A} = \sum_{i\notin A} a_i. \tag{111}$$

The smallest first-order term over all $|A| \le k$ is therefore obtained by removing the k largest a_i, yielding $\sum_{j=k+1}^n a_{(j)}$. Moreover every refined radius satisfies $\rho_{\mathcal{I}_A} \le \rho_0$. Applying Theorem 5 uniformly to all admissible A shows that optimizing the exact gap can change this minimum only by a term bounded in magnitude by $L_m\rho_0^2/2$, which proves (110). □

Equation (110) gives a simple operational rule that is absent from an unweighted information count: a coordinate is valuable when it is both poorly resolved and strongly aligned with the stability gradient. With unequal primitive costs w_i, the first-order design becomes the budgeted selection problem

$$\max_{A\subseteq[n]} \sum_{i\in A} a_i \qquad \text{s.t.} \qquad \sum_{i\in A} w_i \le B, \tag{112}$$

which is the local counterpart of the general frontier (33). Thus the exact fiber-wise benchmark, the passivity directional frontier, and the sensitivity approximation all answer the same design question at different levels of resolution: *which additional information buys the largest reduction in irreducible conservatism under a limited reporting-interface budget?*

9 Conclusion

This paper develops a method-independent framework for understanding conservatism in distributed stability certification from the perspective of information. For a prescribed information structure, the best sound certificate is determined by the worst stability margin among all systems that the reporting interface cannot distinguish. This benchmark separates the total conservatism of a particular certificate into information-induced conservatism and method-induced conservatism, and converts information sufficiency into a property of the information fibers induced by the information structure. In particular, indistinguishable systems with materially different stability margins provide constructive witnesses of a limitation that no alternative certificate using the same information can overcome.

The network–node studies reveal a common mechanism behind several apparently different forms of conservatism. Compressing a network into its spectrum can remove the alignment between network modes and heterogeneous node dynamics; compressing matrix-valued node information into scalar or eigenvalue-only reports removes its alignment with critical interconnection directions; and independently certified local blocks omit cross-node relations altogether. The stability value of information is therefore governed not by the number of reported quantities, but by whether the information preserves the relations along which the stability task varies.

The information–conservatism frontier extends this diagnosis into a reporting-interface-design problem by quantifying the best achievable conservatism under restricted admissible information. Its inverse characterizes the minimum information required to attain a prescribed accuracy, while the witness-pair formulation identifies exactly which task-relevant ambiguities must be resolved. The exact directional frontier for matrix passivity reports and the local sensitivity-based information allocation further demonstrate how this principle can be made constructive. The resulting framework thus provides a systematic basis for deciding where additional complexity is worthwhile: improving the analysis method when the available information is underused, or enriching the reporting interface when the information-limited benchmark itself is restrictive.

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